

Financial Time Series Analysis

Lecture 1

- ▶ Class of 2022 MSCF admitted students may be watching the Time Series lecture on Thursday, March 25. You may wish to implement privacy protections.
- ▶ TA Session, Friday, March 26, 11:00am-12:00pm (EDT)

Outline

- ▶ Syllabus Material
- ▶ Introduction to Time Series
 - ▶ Time Series Topics to be Covered and Reference Material
 - ▶ Stylized Facts
 - ▶ Stationarity, White Noise, Martingale Difference Sequences

Course Topics Time Series Analysis

- ▶ Stylized facts
- ▶ Stationarity
- ▶ Standard Time Series Models
- ▶ Prediction
- ▶ ARCH/GARCH Models
- ▶ Multivariate Time Series Models
 - ▶ Vector autoregressive processes (VAR)
 - ▶ Dickey Fuller test for Stationarity
 - ▶ Cointegration
- ▶ State Space Models and Kalman Filtering

Time Series References: 1

- ▶ Box, G.E.P., Jenkins, G., Reinsel, G. and G.M. Ljung, *"Time Series Analysis: Forecasting and Control, 5th edition, 2015.*
- ▶ Brockwell, P.J. and R. Davis, *Introduction to Time Series and Forecasting, 3rd Edition, 2016.*
- ▶ Chan, N.H., *Time Series: Applications to Finance with R and S-Plus, 2010.*
- ▶ Hamilton, James, *Time Series Analysis, 1994.*
- ▶ Lai, T.L. and Xing, H., *Statistical Models and Methods for Financial Markets, 2008.*
- ▶ McKinney, W., *Python for Data Analysis: Data Wrangling with Pandas, Numpy and IPython, 2nd Edition, 2018.*

Time Series References: 2

- ▶ Ruppert, D. and D. Matteson, *Statistics and Data Analysis for Financial Engineering, 2nd Edition*, 2015.
- ▶ Sargent, T.J. and J. Stachurski, *Quantitative Economics with Python*, 2020. On-line <http://python.quantecon.org> .
- ▶ Tsay, R.S., *An Introduction to the Analysis of Financial Data with R*, 2013.
- ▶ Tsay, R.S., *Analysis of Financial Time Series, 3rd Edition*, 2019.
- ▶ Tsay, R.S. *Multivariate Time Series Analysis with R and Financial Applications*, 2014.
- ▶ Velu, R., Hardy, M. and D. Nehren, *Algorithmic Trading and Quantitative Strategies*, 2020.

Python Packages

- ▶ numpy, scipy, pandas, matplotlib
- ▶ statsmodels
- ▶ quantecon (Sargent and Stachurski text)
- ▶ pmdarima (claims to have functionality of R package `auto.arima`)
- ▶ arch (for ARCH/GARCH)
- ▶ johansen (for Johansen cointegration test)
- ▶ pyflux (time series analysis and prediction)
- ▶ state-space, pykalman
- ▶ See pypi.org, python package index

Time Series In Economics and Finance

- ▶ Time series analysis is a set of models and methodologies to handle non-independent data (univariate or multivariate) with a temporal dependence structure.
- ▶ Time series methods apply in many areas of finance:
 - ▶ Predicting future values of economic and financial time series like stock prices (prediction)
 - ▶ Identify sets of stocks that have a temporal dependence relationship, and use these in a convergence trading and strategy. (see next slides)
 - ▶ Model volatility clustering and use these models in risk management calculations. (GARCH - volatility clustering)

An Example: Convergence Trading: 1

- ▶ Consider two stocks that appear to be moving together (in standard units).
- ▶ If the price of stock 1 increases more than that of stock 2, one can short the relatively higher priced stock and long the lower price stock and wait for the two stock prices to revert to a common level.
- ▶ if this reversion occurs, the profit is the difference in prices at the time of the trade.

An Example: Convergence Trading: 2

- ▶ Such a trade is “zero-cost” and “market neutral”
- ▶ How do we determine that pairs (or baskets) of stocks are “moving together?”
- ▶ **Cointegration** invented by Engle and Granger is a concept to identify if a linear combination of non-stationary time series are stationary. There are statistical procedures to identify cointegrated series.

Gatev et al: Pairs Trading

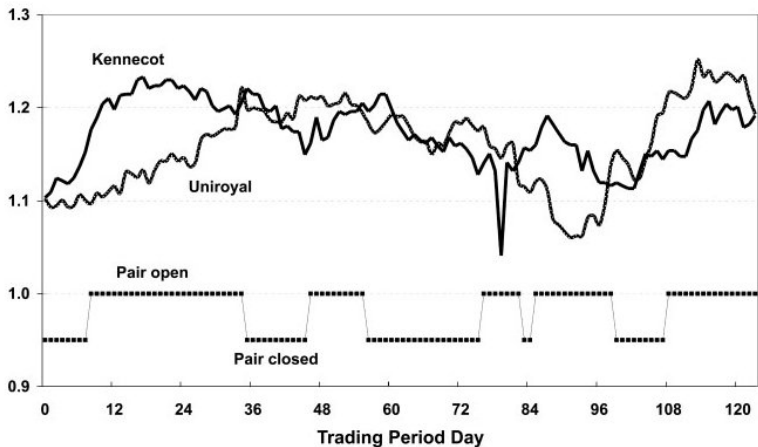


Figure 1
Daily normalized prices: Kennecott and Uniroyal (pair 5)
Trading period August 1963–January 1964.

Empirical Properties of Asset Returns: 1

- ▶ The availability of data and computing power is fundamental to understanding the price behavior of financial assets.
- ▶ Before 1960, there was no central database of security prices.
- ▶ In 1960, CRSP was founded at the University of Chicago, and a major project was the establishment of daily stock price data (open, high, low, close) for all NYSE stocks. This became available in 1962. Later expanded to other markets (AMEX, NASDAQ)

Empirical Properties of Asset Returns: 2

- ▶ In mathematical finance, the basic model of asset price processes is GBM, $S_{t+h} = S_t \exp((r - .5\sigma^2)h + \sigma W(h))$. This dates back to Bachelier in 1900.
- ▶ Calculating the log-returns over a time period of length j , we get $\log(S_{t+h}/S_t) = (r - .5\sigma^2)h + \sigma W(h)$, which has a $N((r - .5\sigma^2)h, \sigma^2 h)$ distribution.
- ▶ Thus we would expect a good model for log-returns would be a normal distribution.
- ▶ CRSP data allowed for a careful study (on a daily time scale) of the behavior of stock prices. Just like the microscope revolutionized biology and the telescope revolutionized astronomy, CRSP data revolutionized quantitative stock price modeling.

Empirical Properties of Asset Returns: 3

- ▶ In 1965, Fama published a major empirical study of the behavior of stock prices. He demonstrated that log-returns have much heavier tails than would be predicted from a normal distribution.
- ▶ Fama erroneously concluded that the log-returns should be modeled using **stable laws**.
- ▶ The stable laws (indexed by $0 < \alpha \leq 2$) (e.g. Cauchy, $\alpha = 1$) have no moments $\geq \alpha$.
- ▶ A 2001 paper by Rama Cont on “Empirical properties of asset returns and stylized facts” presents a number of observations that characterize the general behavior of financial asset prices.

Empirical Properties of Asset Returns: 4

- ▶ The Cont paper on stylized facts shows that the i.i.d. normal model is most often inappropriate for dealing with financial data. Some of the key departures from standard statistical models include
 - ▶ Lack of stability in long-run behavior for raw financial variables
 - ▶ Heavy tails in returns (but higher moments exist)
 - ▶ Temporal dependencies
 - ▶ Volatility clustering (GARCH models)

Stylized Facts: 1

Rama Cont, “Empirical properties of asset returns: stylized facts and statistical issues,” *Quantitative Finance*, **1**, 2001, 223-236. These are presented here, out of order.

1) Absence of autocorrelations:

“(linear) autocorrelations of asset returns are often insignificant, except for very small intraday time scales for which microstructure effects come into play.”

We will define the autocovariance and autocorrelation function later. However, a plot of the autocovariance function should show a spike at 0 for the variance of the process, possibly a significant, but smaller spike at lag 1 and no significant spikes beyond 1.

Stylized Facts: 2

2) Heavy Tails:

“the (unconditional) distribution of returns seems to display a power-law or Pareto-like tail, with a tail index which is finite, higher than two and less than five for most data sets studied. In particular this excludes stable laws with infinite variance and the normal distribution. However the precise form of the tails is difficult to determine.”

3) Gains/loss asymmetry:

“one observes large drawdowns in stock prices and stock index values but not equally large upward movements. (this property is not true for FX where there is symmetry).”

Stylized Facts: 3

4) Aggregational Gaussianity:

“as one increases the time scale over which returns are calculated, their distribution looks more like a normal distribution. The shape is not the same at different time scales.”

5) Intermittency:

“returns display, at any time scale, a high degree of variability. This is quantified by the presence of irregular bursts in time series of a wide variety of volatility estimators.”

Stylized Facts: 4

6) Volatility clustering:

“different measures of volatility display a positive autocorrelation over several days, which quantifies the fact that high-volatility events tend to cluster in time.”

7) Conditional heavy tails:

“even after correcting returns for volatility clustering (e.g. via GARCH-type models), the residual time series still exhibit heavy tails. However, the tails are less heavy than in the unconditional distribution of returns.”

8) Slow decay of autocorrelation in absolute returns:

“the autocorrelation function of absolute returns decays slowly as a function of the time lag, roughly as a power law with an exponent $\beta \in [0.2, 0.4]$. This is sometimes interpreted as a sign of long-range dependence.

Stylized Facts: 5

9) Leverage effect:

"most measures of volatility of an asset are negatively correlated with the returns of that asset."

P668 of Hamilton's text "A number of researchers have found evidence of asymmetry in stock price behavior - negative surprises seem to increase volatility more than positive surprises. Since a lower stock price reduces the value of equity relative to corporate debt, a sharp decline in stock prices increases corporate leverage and could thus increase the risk of holding stocks. For this reason, the apparent finding ... is called the **leverage effect**."

Stylized Facts: 6

10) Volume/volatility correlation:

"trading volume is correlated with all measures of volatility."

11) Asymmetry in time scales:

"coarse-grained measures of volatility predict fine-scale volatility better than the other way around."

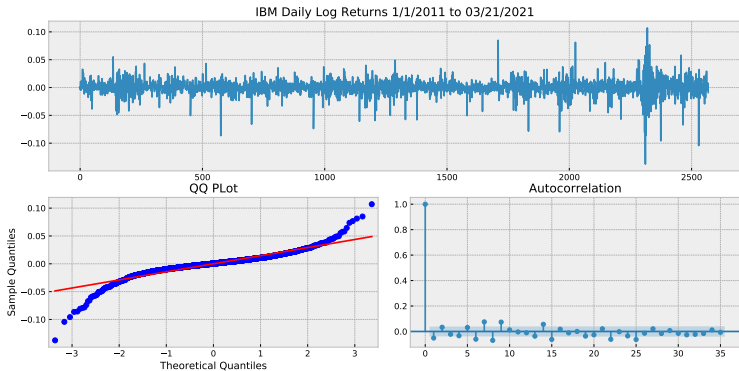
Log>Returns: 1

- ▶ The standard central limit theorem indicates that sums of i.i.d. random variables with finite variance, when properly normalized converge weakly to a normal distribution.
- ▶ There are many variations, and roughly the identical distribution part can be weakened as long as some of the summands don't predominate (i.e. an elephant among many ants).
- ▶ The independence assumption can be weakened as long as the dependence dies off quickly.

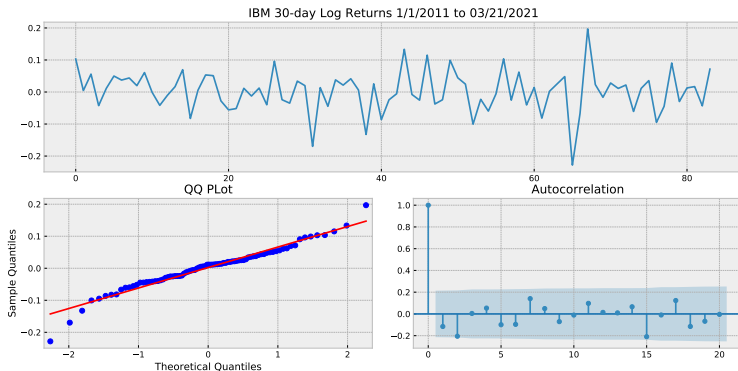
Log>Returns: 2

- ▶ Consider log-returns, $\log(S_{t+1}/S_t)$ and $\log(S_{t+N}/S_t)$
- ▶ Note $\log(S_{t+N}/S_t) = \log\left(\frac{S_{t+N}}{S_{t+N-1}} \cdots \frac{S_{t+1}}{S_t}\right) = \sum_{i=1}^N \log\left(\frac{S_{t+i}}{S_{t+i-1}}\right)$
- ▶ Thus log-returns should be normal over longer time horizons provided:
 - ▶ Variances are finite
 - ▶ The autocorrelation is small
 - ▶ No terms dominate the sum.

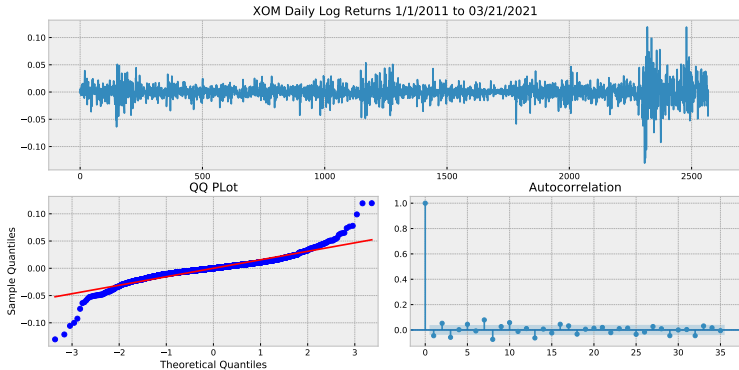
IBM Daily Log Returns 1/1/2011-3/20/2021



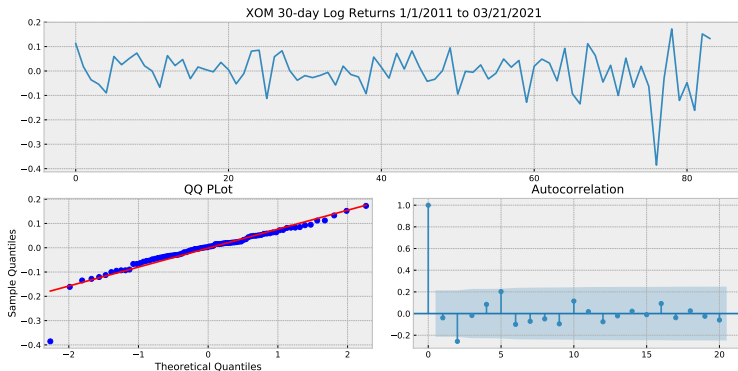
IBM 30-Day Log Returns 1/1/2011-3/20/2021



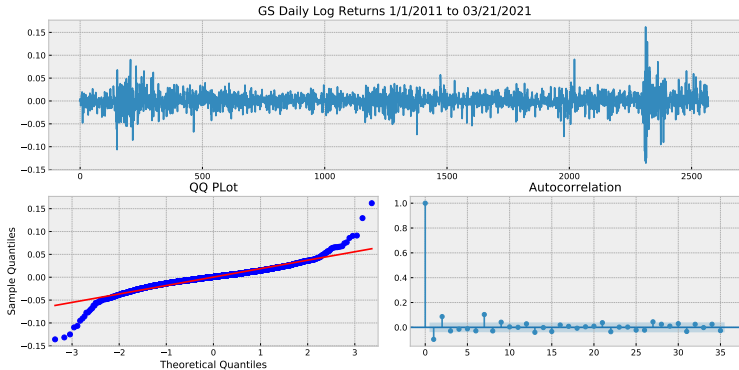
XOM Daily Log Returns 1/1/2011-3/20/2021



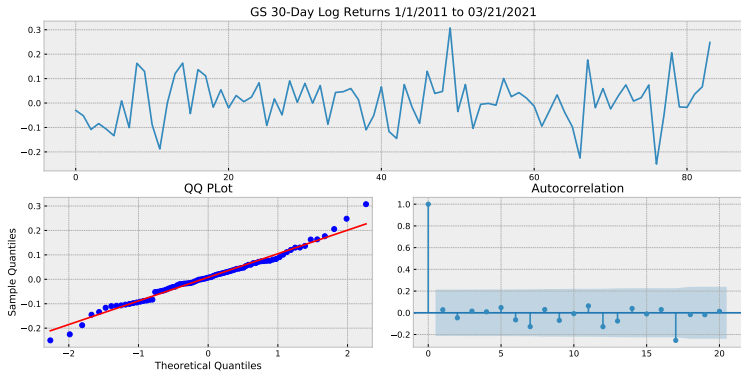
XOM 30-Day Log Returns 1/1/2011-3/20/2021



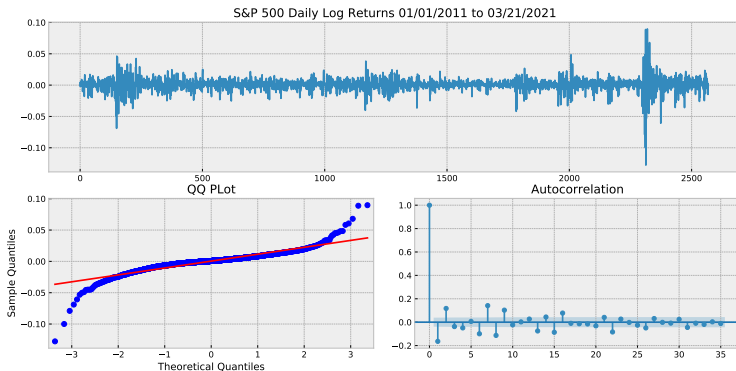
GS Daily Log Returns 1/1/2011-3/20/2021



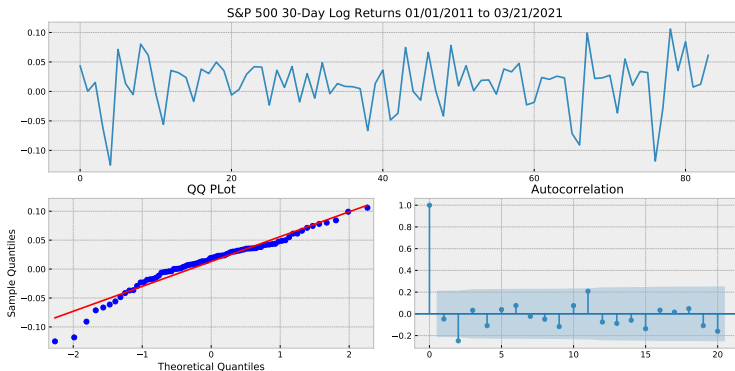
GS 30-Day Log Returns 1/1/2011-3/20/2021



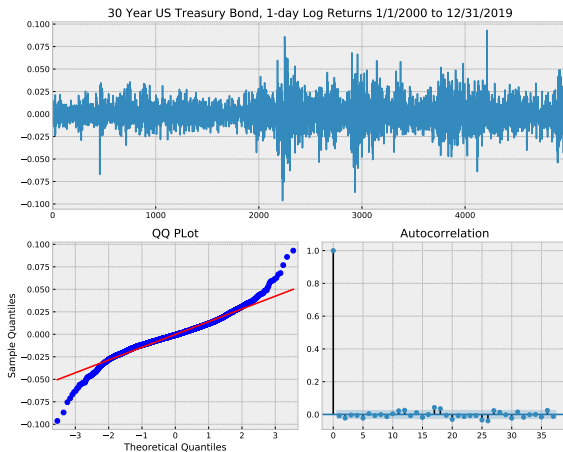
S&P 500 Daily Log Returns 1/1/2011-3/20/2021



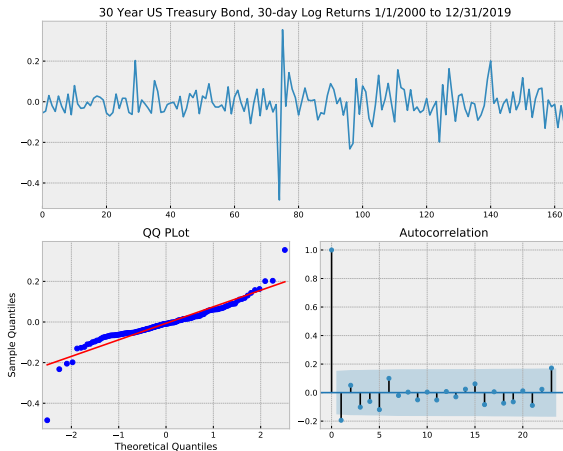
S&P 500 30-Day Log Returns 1/1/2011-3/20/2021



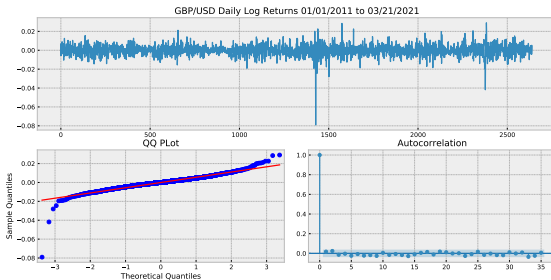
US 30-Year Treasury Bond Daily Returns 1/1/2000-12/31/2019



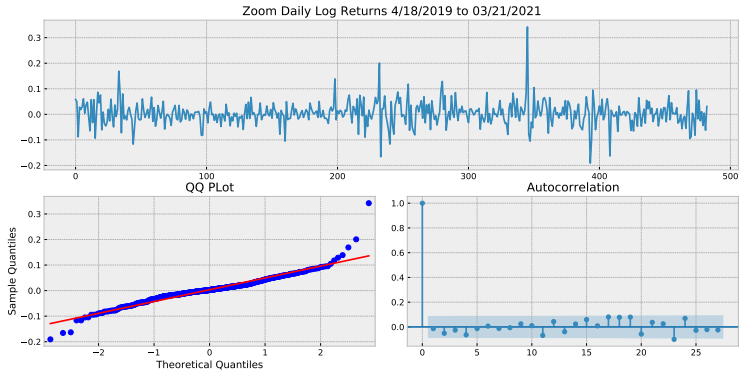
US 30-Year Treasury Bond 30-Day Returns 1/1/2000-12/31/2019



GBP/USD Daily Log>Returns 1/1/2011-3/21/2021



Zoom Video Daily Returns 4/16/2019-3/20/2021



Time Series Models/Graphics: 1

```
import os
import sys
import pandas as pd
import pandas_datareader as pdr
import numpy as np
import statsmodels as smm
import statsmodels.formula.api as smf
import statsmodels.tsa.api as smt
import statsmodels.api as sm
import scipy.stats as scs
import matplotlib.pyplot as plt
import matplotlib as mp
import seaborn as sns
```

Time Series Models/Graphics: 2

```
def tsplot(y, lags=None, figsize = (10,8), style = 'bmh'):
    if not isinstance(y, pd.Series):
        y = pd.Series(y)
    with plt.style.context(style):
        fig = plt.figure(figsize=figsize)
        layout = (2,2)
        ts_ax = plt.subplot2grid(layout, (0,0), colspan = 2)
        acf_ax = plt.subplot2grid(layout, (1,1))
        qq_ax = plt.subplot2grid(layout, (1,0))
        y.plot(ax = ts_ax)
        ts_ax.set_title('Insert Title')
        smt.graphics.plot_acf(y, lags = lags, ax=acf_ax,
            alpha = 0.05)
        sm.qqplot(y, line = 's', ax=qq_ax)
        qq_ax.set_title('QQ PLOT')
        plt.tight_layout()

    return
```

Time Series Models/Graphics: 3

```
start = '2011-01-01'
end = '2021-03-21'
price = pdr.get_data_yahoo('gs',start,end)
adjclose = price_data['Adj Close']
shf = 1
logret = np.log(adjclose/adjclose.shift(shf)).dropna()
L = len(logret)
LL = int(L/shf)
logret_shf = np.zeros(LL)
for i in range(int(L/shf)):
    logret_shf[i] = logret[i*shf]
.
tsplot(logret_shf)
plt.show()
```

STATIONARITY

Stationarity: 1

- ▶ Consider a time series $\{X_t\}$ for $t = 0, 1, 2, \dots$ or $t = 0, \pm 1, \pm 2, \dots$. This time series is called
 - ▶ **Strictly Stationary** if for all $n \geq 1$, $t_1 < t_2 < \dots < t_n$, and $i = 1, 2, \dots$ the joint distribution of $(X_{t_1}, X_{t_2}, \dots, X_{t_n})$ is the same as $(X_{t_1+i}, X_{t_2+i}, \dots, X_{t_n+i})$.
 - ▶ **Weakly Stationary** if for all $t_1 < t_2$ and $i = 1, 2, \dots$, $\text{Cov}(X_{t_1}, X_{t_2}) = \text{Cov}(X_{t_1+i}, X_{t_2+i})$.
- ▶ We will always use the wider or less restrictive version of stationarity, weak stationarity also known as covariance stationarity or second order stationarity.

Stationarity: 2

- ▶ A covariance stationary process $\{X_t, t \in \mathcal{Z}\}$ satisfies:
 - ▶ $E(X_t) = \mu, t \in \mathcal{Z},$
 - ▶ $Cov(X_s, X_t) = \gamma(s, t) = \gamma(s + k, t + k), s, t, k \in \mathcal{Z}$
- ▶ Since $\gamma(s, t) = \gamma(0, t - s) = \gamma(t - s, 0)$, the function γ is actually a function of only one variable, $|t - s|$ the **lag**.
- ▶ We define the **autocovariance function**, $\gamma(h) = \gamma(h, 0)$
- ▶ Note that $\gamma(0) = Var(X_t)$ and the autocorrelation function, $\rho(h)$, is defined by $\rho(h) = \gamma(h)/\gamma(0)$.

Stationarity: 3

- ▶ The backbone of much of basic statistics is the i.i.d. process. In time series analysis, the corresponding backbone is the **White Noise Process**
- ▶ A **white noise process** is a weakly stationary process with constant mean 0, and autocovariance function $\gamma(h) = \sigma^2$ if $h = 0$ and 0 if $h \neq 0$.
- ▶ A white noise process is denoted by $WN(0, \sigma^2)$.
- ▶ The autocorrelation function for a white noise process, $\rho(h) = 1$ if $h = 0$ and 0 if $h \neq 0$